

■ 3.7.11 Advanced Topic: Singular Value Decomposition

<code>SingularValues[m]</code>	find the singular value decomposition of a numerical matrix
<code>PseudoInverse[m]</code>	compute the pseudoinverse of a numerical matrix

Some advanced numerical matrix operations.

Singular value decomposition is an important element of many numerical matrix algorithms. The basic idea is to write any matrix $\mathbf{m}$ in the form $\mathbf{u}^T \mathbf{m}_D \mathbf{v}$, where $\mathbf{m}_D$ is a diagonal matrix, and $\mathbf{u}$ and $\mathbf{v}$ are respectively row and column orthogonal matrices. The function `SingularValues[m]` returns a list containing the matrix $\mathbf{u}$, the list of diagonal elements of $\mathbf{m}_D$, and the matrix $\mathbf{v}$.

The diagonal elements of $\mathbf{m}_D$ are known as the *singular values* of the matrix $\mathbf{m}$. One interpretation of the singular values is as follows. If you take a sphere in n -dimensional space, and multiply each vector in it by an $m \times n$ matrix $\mathbf{m}$, you will get an ellipsoid in m -dimensional space. The singular values give the lengths of the principal axes of the ellipsoid. If the matrix $\mathbf{m}$ is singular in some way, this will be reflected in the shape of the ellipsoid. In fact, the ratio of the largest singular value of a matrix to the smallest one gives the *condition number* of the matrix, which determines, for example, the accuracy of numerical matrix inverses.

In general, very small singular values are numerically meaningless. `SingularValues` removes any singular values that are smaller than a certain *tolerance* multiplied by the largest singular value. The option `Tolerance` specifies the tolerance to use. The default tolerance is ten times the precision of the numbers in the original matrix $\mathbf{m}$.

Here is the singular value decomposition for a non-singular 2×2 matrix.

```
In[1]:= SingularValues[N[{{1, 3}, {-4, 3}}]]
Out[1]= {{{-0.957092, 0.289784}, {-0.289784, -0.957092}},
          {2.91309, 5.14916},
          {{-0.726454, -0.687215}, {0.687215, -0.726454}}}
```

Here is a convenient way to pick out the pieces of the result.

```
In[2]:= {u, md, v} = %
Out[2]= {{{-0.957092, 0.289784}, {-0.289784, -0.957092}},
          {2.91309, 5.14916},
          {{-0.726454, -0.687215}, {0.687215, -0.726454}}}
```

This gives something close to the original matrix.

```
In[3]:= Transpose[u] . DiagonalMatrix[md] . v
Out[3]= {{1., 3.}, {-4., 3.}}
```

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Here is the singular value decomposition of a singular matrix. Only one singular value is given.

```
In[4]:= SingularValues[N[{{1, 2}, {1, 2}}]]
Out[4]= {{0.707107, 0.707107}}, {3.16228},
        {{0.447214, 0.894427}}}
```

The standard definition for the inverse of a matrix fails if the matrix is not square. Using singular value decomposition, however, it is possible to define a *pseudoinverse* even for non-square matrices, or for singular square ones. The pseudoinverse is defined in terms of the objects $\mathbf{u}$, $\mathbf{m}_D$ and $\mathbf{v}$ as $\mathbf{m}^{(-1)} = \mathbf{v}^T \mathbf{m}_D^{-1} \mathbf{u}$. The pseudoinverse has the property that the sum of the squares of all the entries in $\mathbf{m} \cdot \mathbf{m}^{(-1)} - \mathbf{I}$, where $\mathbf{I}$ is an identity matrix, is minimized. The pseudoinverse found in this way is important, for example, in carrying out fits to numerical data. The pseudoinverse is sometimes known as the generalized inverse, or the Moore-Penrose inverse.